Kinetic Theory of gases

Gases: It is a collection of molecules which are moving rapidly in random direction.

- → Ideal gas = It follows some assumptions.
- → Real gas

Assumptions of ideal gas:

→ Number of gaseous molecules in container is very large.

→ Gaseous molecules are identical, elastic very small solid sphere

$e = 1 \Rightarrow$ No loss in collision

→ Volume of gas = Volume of container

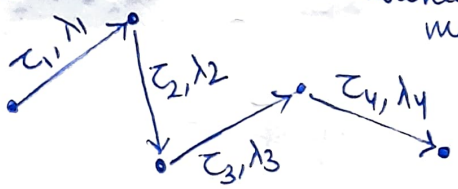
→ No intermolecular forces b/w gaseous molecules.
⇒ No potential energy.

→ All the molecules are moving in every possible random direction without giving preference to any direction.

→ If container is at rest, the avg. velocity of gaseous molecules is zero.

→ Gaseous molecules will collide with ① wall of container ② other molecules } elastic collision

This type of motion is called brownian motion



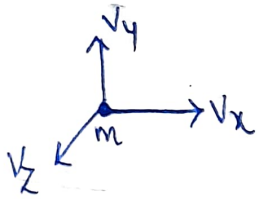
τ = Relaxation time = Time interval b/w two collision

λ = free path = distance b/w two molecules.

$$\tau_{mean} = \frac{\tau_1 + \tau_2 + \tau_3 + \tau_4 + \dots}{n}$$

$$\lambda_{mean} = \frac{\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \dots}{n}$$

→



$$\vec{V} = V_x\hat{i} + V_y\hat{j} + V_z\hat{k}$$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}m\{V_x^2 + V_y^2 + V_z\}^2$$

If $V_x = V_y = V_z \Rightarrow$ $\boxed{V_x^2 = \frac{V^2}{3}}$

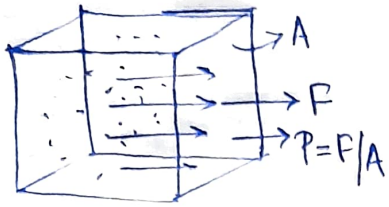
Energy of gas = $KE + \cancel{PE}^{\to 0}$

$= \sum\left(\frac{1}{2}mv^2\right)$ = internal energy of gas

→ We are neglecting gravity ⇒ we can assume that density of gas is uniform

Pressure of gas:

→ Force applied by colliding gas molecules on walls of container per unit area is defined as pressure of gas.



→ Unit = N/m^2 = Pascal
→ 1atm = 1.01×10^5 Pa
1bar = 10^5 Pa
→ Dimension = $[M L^{-1} T^{-2}]$
→ Scalar

Volume of gas: It is volume of container.

No. of moles of gas:

$$n = \frac{\text{no. of molecules of gas}}{\text{Avagadro No.}} = \frac{\text{mass of gas}}{\text{molar mass of gas}} = \frac{m}{M}$$

$N_A = 6.023 \times 10^{23}$

$$n = \frac{N}{N_A}$$

Q Find n of 480 gm O_2

$$n = \frac{m}{M} = \frac{480\ gm}{32\ gm/mol} = 15 \text{ moles}$$

Temperature of gas:



0K = −273°C ⇒ Super condensed state
ie KE = 0

Now, as T↑ ⇒ vibrations↑ ⇒ KE↑

* High T means high random speeds of gaseous molecules and hence high KE of gas.

Boyle's law ⇒ At const T, $P \propto \frac{1}{V}$

Charles law ⇒ At const P, $V \propto T$

Ideal gas equation:

$$\boxed{PV = nRT}$$

P = pressure of gas
V = volume of gas
n = no. of moles
R = gas const.
T = Temp. in Kelvin {K = °C + 273}

$$\boxed{R = 8.314\ J/mol\text{-}K}$$

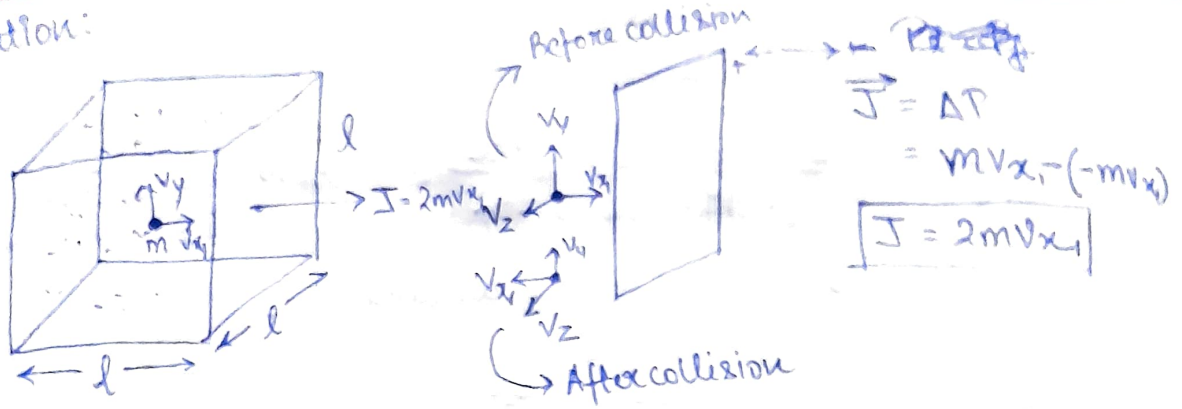
$$\boxed{R = \frac{25}{3}\ J/mol\text{-}K}$$

$$\frac{R}{N_A} = \frac{8.314}{6.023 \times 10^{23}}$$

$R = 1.38 \times 10^{-23}$

R = Boltzman constant

Derivation:



$\vec{I} = \Delta P$

$= mv_{x_1} - (-mv_{x_1})$

$\boxed{I = 2mv_{x_1}}$

Time interval in which particle applies impulse on wall $= \frac{2l}{V_{x_1}}$

$F_1 = \frac{2mV_{x_1}^2}{2l} = \frac{mV_{x_1}^2}{l}$

$F = \Sigma F_1 = \frac{Nm}{l} \frac{\{V_{x_1}^2 + V_{x_2}^2 + V_{x_3}^2 + V_{x_4}^2 + \cdots V_{x_n}^2\}}{N}$ { Multiplying and dividing by N = no. of molecules }

Now, Nm = mass of gas $= nM$

$F = \frac{nM}{l} \langle V_x^2 \rangle$ { $\langle V_i^2 \rangle$ = Mean of V_x^2 }

Also, $\langle V_x^2 \rangle = \langle V_y^2 \rangle = \langle V_z \rangle^2$

$\Rightarrow \langle V_x^2 \rangle + \langle V_y^2 \rangle + \langle V_z \rangle^2 = \langle V^2 \rangle \Rightarrow \langle V_x^2 \rangle = \frac{\langle V^2 \rangle}{3}$

$\boxed{F = \frac{nM}{l} \frac{\langle V^2 \rangle}{3}} \Rightarrow P = \frac{F}{A} = \frac{(\text{mass of gas})}{l^3} \frac{\langle V^2 \rangle}{3}$

$P = \frac{\rho}{3} V_{Rms}^2 \Rightarrow (V_{Rms})^2 = \frac{3P}{\rho} \Rightarrow * \boxed{V_{rms} = \sqrt{\frac{3P}{\rho}}}$

So,

$\rightarrow V_1$ $\uparrow V_3$
$\leftarrow V_2$ $\downarrow V_4$
$\leftarrow V_5$ $\downarrow V_n$

Avg speed $= \frac{V_1 + V_2 + V_3 + \cdots + V_n}{n}$

Root mean square speed $= \sqrt{\frac{V_1^2 + V_2^2 + V_3^2 + \cdots + V_n^2}{n}} = \sqrt{\frac{3P}{\rho}}$

Now, $PV = nRT$

$PV = \frac{m}{M} RT$

$PM = \frac{m}{V} RT$

$\boxed{PM = \rho RT}$

$\boxed{\frac{P}{\rho} = \frac{RT}{M}}$ $\therefore \boxed{V_{rms} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3RT}{M}}}$

Q $P = 3\times10^6 N/m^2$ Find RMS speed of gas
$\rho = 1.44 kg/m^3$

$$V_{Rms} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3\times3\times10^6}{1.44}} = \frac{3\times10^3}{1.2} = 2.5\times10^3 m/s = 2.5 km/s$$

Q

O_2 gas
$T = 27^\circ C$

Find V_{rms} of O_2 gas in container.

$$V_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3\times\frac{25}{3}\times300}{32\times10^{-3}}} = \sqrt{\frac{25\times3\times10^5}{32}} = 484\ m/s$$

Q Find volume of 1 mole of gas at STP

STP $\Rightarrow T = 0^\circ C$

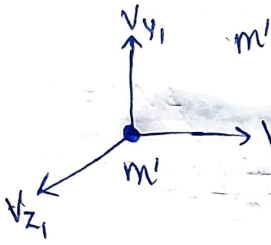
$P = 1.013\times10^5 N/m^3$

$PV = nRT$

$$V = \frac{nRT}{P} = \frac{1\times8.314\times273}{1.013\times10^5} = 2240\times10^{-5} m^3 = 22.4\times10^{-3} m^3 = 22.4 \text{ litres}$$

$(1 lt = 10^{-3} m^3)$

Translational Kinetic Energy of gas:



m' = mass of 1 molecule of gas

m = mass of gas.

$$KE_1 = \frac{1}{2}m\left(V_{x_1}^2 + V_{y_1}^2 + V_{z_1}^2\right)$$

$$KE_2 = \frac{1}{2}m\left(V_{x_2}^2 + V_{y_2}^2 + V_{z_2}^2\right)$$

$$\vdots$$

$$KE_n = \frac{1}{2}m\left(V_{x_n}^2 + V_{y_n}^2 + V_{z_n}^2\right)$$

$$KE = \frac{1}{2}m'\left(\sum V_x^2 + \sum V_y^2 + \sum V_z^2\right)$$

Multiply and divide by N = no. of molecules

$$KE = \frac{1}{2}m'N\left(\frac{\sum V_x^2}{N} + \frac{\sum V_y^2}{N} + \frac{\sum V_z^2}{N}\right) = \frac{1}{2}m V_{rms}^2$$

$$\boxed{TKE = \frac{1}{2}m V_{rms}^2}$$

$$TKE = \frac{1}{2}m\left(\frac{3P}{\rho}\right) = \frac{1}{2}\frac{m}{m}3PV = \frac{3}{2}PV$$

$$\therefore \boxed{TKE = \frac{3}{2}PV = \frac{3}{2}nRT}$$

Q Given $P = 2 atm$ Find TKE of gas

$V = 5 lt$

$$TKE = \frac{3}{2}PV = \frac{3}{2}\times2\times1.013\times10^5\times5\times10^{-3} = 15\times1.013\times10^2$$

Q

$37^\circ C$
16gm He

Find TKE.

$$n = \frac{16 gm}{4 gm/mol} = 4 mol$$

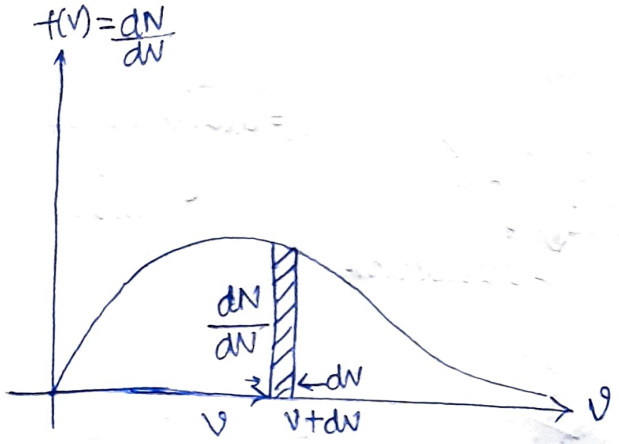
$$TKE = \frac{3}{2}nRT = \frac{3}{2}\times4\times\frac{25}{3}\times310$$

$$= 50\times310$$

$$= 15500 J$$

Maxwell Boltzman Distribution law:

→ Gave an idea about speed of gaseous molecules in a closed container.
→ Gaseous molecules will have random speeds, varying from 0 to ∞.
→ At high T, more molecules will have higher speeds.
→ At less T, more molecules will have lesser speeds.



$$f(v) = \frac{dN}{dv} = 4\pi N v^2 \left(\frac{m}{2\pi KT}\right)^{3/2} e^{-\frac{mv}{2KT}}$$

N = total no. of gaseous molecules
m = mass of 1 molecule = M/N_A
K = Boltzman const = R/N_A
T = temp. in Kelvin

Area of strip = $\frac{dN}{dv} \cdot dv = dN$ = No. of gaseous molecules having speed b/w v to v+dv

Area of curve = $\int \frac{dN}{dv} \cdot dv = \int dN$

= N = no. of gaseous molecules.

Now, we can also write f(v) as:-

$$f(v) = \frac{dN}{dv} = 4\pi N v^2 \left(\frac{M}{2\pi RT}\right)^{3/2} e^{-\frac{Mv^2}{2RT}}$$

① Most probable speed = speed which most of gaseous molecules have. (v_{mp})

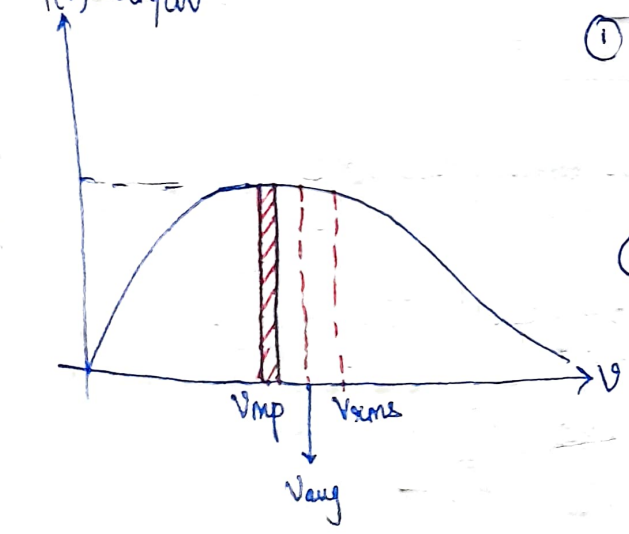
② Avg speed = v_{avg} = mean speed ⇒

$$v_{avg} = \frac{\int v\,dN}{\int dN}$$

③ Root mean square speed = v_{rms}

$$v_{rms} = \sqrt{\frac{\int v^2\,dN}{\int dN}}$$

Now,



$$f(v) = 4\pi N v^2 \left(\frac{m}{2\pi RT}\right)^{3/2} e^{-\frac{mv}{2RT}}$$

① $v_{mp} \Rightarrow f'(v) = 0$

$$v_{mp} = \sqrt{\frac{2kT}{m}} = \sqrt{\frac{2RT}{M}}$$

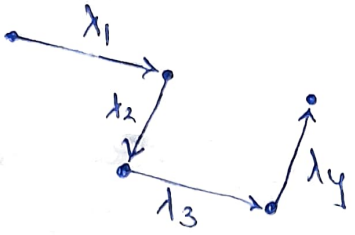
② $$v_{avg} = \frac{\int v\,dN}{\int dN} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8kT}{\pi m}}$$

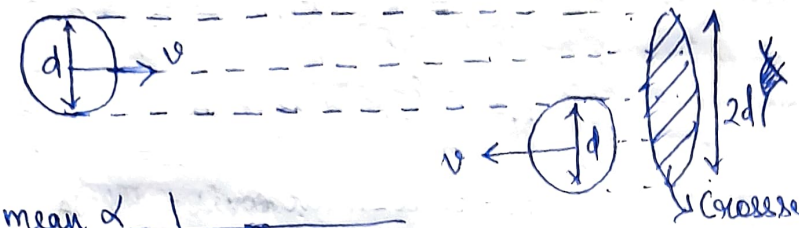
③ $$v_{rms} = \sqrt{\frac{\int v^2\,dN}{\int dN}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3kT}{m}}$$

Mean free path:

→ Distance travelled by gaseous molecules b/w 2 consecutive collisions is known as free path.

→ Average of these free paths is mean free path





d = diameter of gaseous molecule

$$\lambda_{mean} \propto \frac{1}{\text{area of cross section}}$$

$$\therefore \lambda_{mean} \propto \frac{1}{\pi d^2}$$

Also, $\lambda_{mean} \propto \frac{1}{N_v}$

N_v = particle density = $\frac{\text{No. of gaseous molecules}}{\text{volume}}$

$$\boxed{\lambda_{mean} = \frac{1}{\sqrt{2}\pi d^2 N_v}}$$

$$\boxed{\lambda_{mean} = \frac{V}{\sqrt{2}\pi d^2 N}}$$

Also, $N = nN_A$

$$\Rightarrow \lambda_{mean} = \frac{V}{\sqrt{2}\pi d^2 n N_A}$$

Also, $PV = nRT$

$$\lambda_{mean} = \frac{VRT}{\sqrt{2}\pi d^2 PV N_A} = \frac{RT}{\sqrt{2}\pi d^2 P N_A}$$

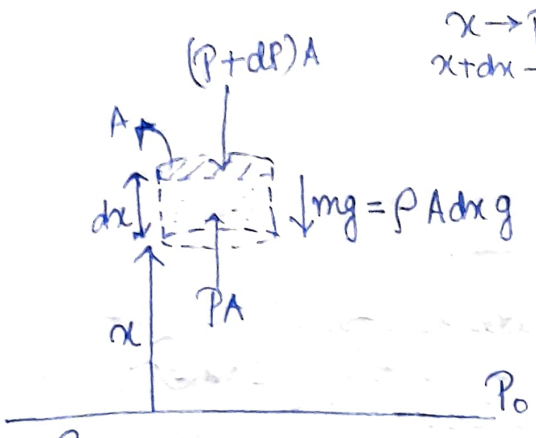
Now, time interval between two consecutive collisions by gaseous molecules is known as relaxation time (τ)
Average of these is mean relaxation time.

$$\boxed{\tau_{mean} = \frac{\lambda_{mean}}{v_{rms}}}$$

Note: We know, $\boxed{PM = \rho RT}$

In liquids, density doesn't change much with pressure.
In gases, on increasing pressure, density increases.

** Q Given pressure at surface is P_0. Also given M, T for gas.
Accn due to gravity = g.
Find pressure at height h from ground.



$PA = (P+dP)A + (\rho A dx)g$

$$\boxed{dP = -\rho g\, dx}$$

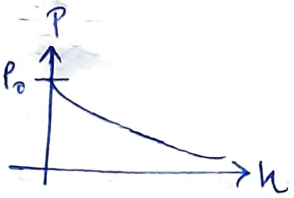
Now, $PM = \rho RT$

$$\rho = \frac{PM}{RT}$$

$$dP = -\frac{PM}{RT} g\, dx$$

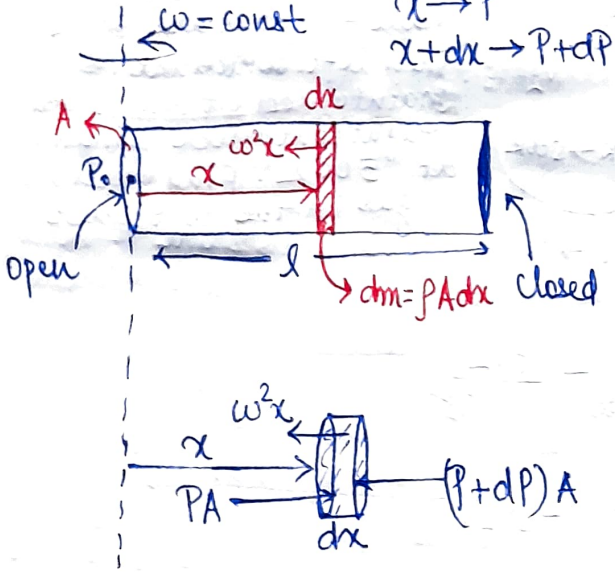
$$= \int_{P_0}^{P} \frac{dP}{P} = -\frac{Mg}{RT}\int_0^h dx \Rightarrow (\ln P)_{P_0}^{P} = -\frac{Mg}{RT}(x)_0^h$$

$$= \ln\left(\frac{P}{P_0}\right) = -\frac{Mgh}{RT} \Rightarrow \boxed{P = P_0 e^{-\frac{Mgh}{RT}}}$$



***

Q No gravity



If pressure at open end is P_0, then find pressure at closed end.
Given:- M, T of gas.

Applying NL:

$$(P+dP)A - PA = (\rho A dx)\omega^2 x$$

$$\boxed{dP = \rho\omega^2 x\, dx}$$

Now, $PM = \rho RT$

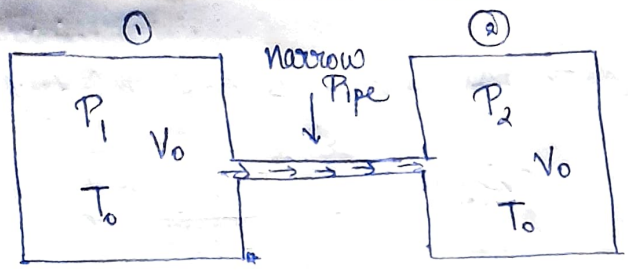
$$\rho = \frac{PM}{RT}$$

$$dP = \frac{PM}{RT}\omega^2 x\, dx$$

$$= \int_{P_0}^{P} \frac{dP}{P} = \frac{M\omega^2}{RT}\int_0^l x\, dx \Rightarrow (\ln P)_{P_0}^{P} = \frac{M\omega^2}{RT}\left(\frac{x^2}{2}\right)_0^l$$

$$= \ln\frac{P}{P_0} = \frac{M\omega^2}{RT}\left(\frac{l^2}{2}\right) \Rightarrow \boxed{P = P_0 e^{\frac{M\omega^2 l^2}{2RT}}}$$

Q



Find final pressure in each container ($P_1 > P_2$)

As gas moves from high P to low P, gas here goes from P_1 to P_2. Due to this no. of moles of gas in ① ↓ and no. of moles of gas in ② ↑. But the total no. of moles of system is conserved.

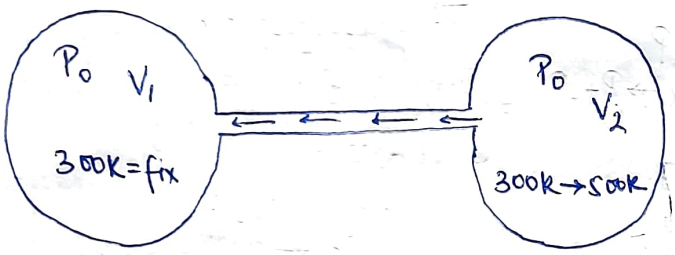
Let in ①, $P_1 \rightarrow P$ and in ②, $P_2 \rightarrow P$

Moles conservation:

$$\frac{P_1 V_0}{RT} + \frac{P_2 V_0}{RT} = \frac{PV_0}{RT} + \frac{PV_0}{RT} \qquad \{PV = nRT,\ n = \frac{PV}{RT}\}$$

$$\boxed{P = \frac{P_1 + P_2}{2}}$$

Q



Let $P_0 \rightarrow P$ Let $P_0 \rightarrow P$

Initially both containers have temperature 300K. Then left container is maintained at 300K and right container is maintained at 500K. Find pressure of gas in the container.

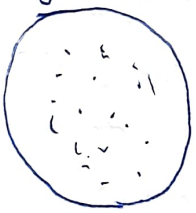
Moles conservation:

$$\frac{P_0 V_1}{R \times 300} + \frac{P_0 V_2}{R \times 300} = \frac{PV_1}{R \times 300} + \frac{PV_2}{R \times 500}$$

$$= \frac{P_0(V_1 + V_2)}{3} = \frac{P}{15}(5V_1 + 3V_2)$$

$$= \boxed{P = \frac{5P_0(V_1 + V_2)}{(5V_1 + 3V_2)}}$$

Q Rigid container (volume fix)



On increasing temperature of gas by 1°C, pressure increases by 1%. Find actual temp of gas.

$P \rightarrow P + \frac{1}{100}P$

$\Rightarrow P \rightarrow 1.01P$

$T \rightarrow T + 1$

$PV = nRT$

$\frac{P}{T} = \text{const}$

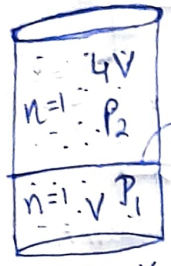
$= \frac{P}{T} = \frac{1.01P}{T+1} \Rightarrow PT + P \times 1 = 1.01PT \Rightarrow T+1 = 1.01T \Rightarrow 0.01T = 1$

$\boxed{T = 100K}$

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Q Find the temperature at which ratio of volumes of two compartments becomes 2:1.

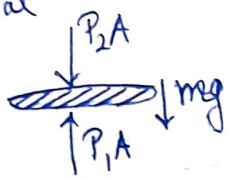
Vertical



freely moving piston eqm

T = 300K

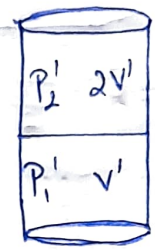
Initial



$P_2A + mg = P_1A$

$\boxed{P_1 - P_2 = \frac{mg}{A}}$

Finally



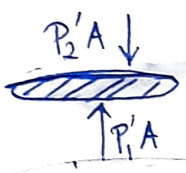
T

Volume should be same before & after

$5V = 3V' \quad -(1)$

$n = \frac{PV}{RT}$

Final



$\boxed{P_1' - P_2' = \frac{mg}{A}}$

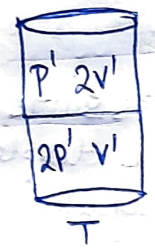
$\frac{P_1' V'}{RT} = \frac{P_2' (2V')}{RT}$

$P_1' = 2P_2' \quad -(2)$

Pressure difference in both cases is same

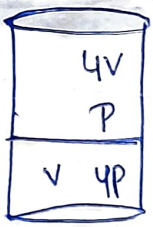
So,

Final



T

Initial



T = 300K

As pressure difference is same,

$4P - P = 2P' - P'$

$\boxed{3P = P'}$

Now, moles conservation

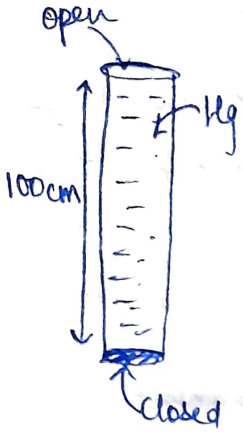
$$\frac{P(4V)}{R \times 300} = \frac{P' \times 2V'}{RT}$$

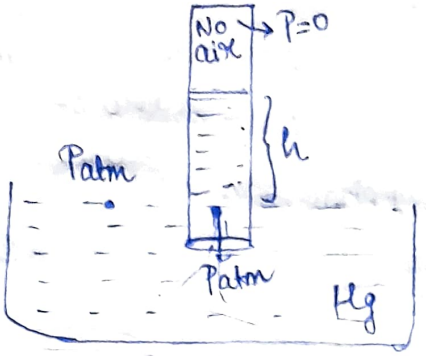
$$\frac{24PV}{300} = \frac{3P \times 2 \times 5V}{3T}$$

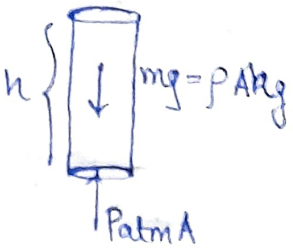
$$T = \frac{300 \times 5}{2} = 750K$$

Barometer:

→ It is used to find atmospheric pressure.







$\rho_{Hg} = 13600\ kg/m^3$

$P_{atm} A = \rho A h g$

$P_{atm} = \rho g h$

At sea level h is observed to be 76cm = 0.76m

$P_{atm} = 13600 \times 9.8 \times 0.76 = 1.01 \times 10^5\ N/m^2$

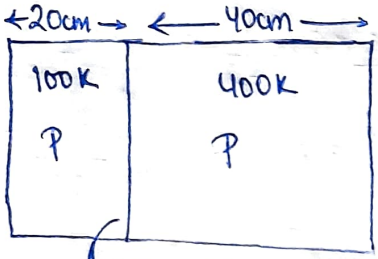
$\therefore P_{atm} = 1.01 \times 10^5\ N/m^2 = 1\ atm$

$= 76\ cm\ of\ Hg = 760\ mm\ of\ Hg$

$= 760\ torr$

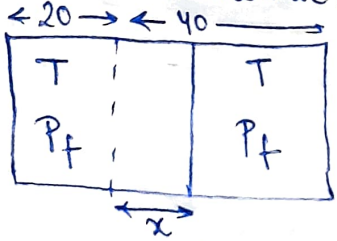
1 torr = 1mm of Hg

Q



Initially pressure is same of both the gases. Find the displacement of piston till it again achieve eq^m

Concept: Heat will flow from high temp to lower temp. 100K increase and 400K decreases. Due to ↑ in T, molecules vibrate faster and applies more pressure which make the piston move. Now, as piston moves, V↑ ⇒ P↓. So finally P should be same in both the compartment



Moles conservation:

$$\frac{P(A)(20cm)}{100} = \frac{P_f(A)(20+x)}{T} \quad -(1)$$

$$\frac{P(A)(40cm)}{400} = \frac{P_f A(40-x)}{T} \quad -(2)$$

Divide (1) and (2)

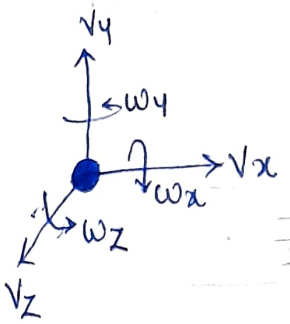
$$\frac{20 \times 400}{100 \times 40} = \frac{20+x}{40-x}$$

$$\boxed{x = 20\ cm}$$

Degree of freedom (f):

→ No. of terms in energy of a gaseous molecule
(Kinetic energy)

① Monoatomic: Ex He, Ne, Ar



$\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$

$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$

Translational Kinetic Energy $= \frac{1}{2}mv^2$

$= \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2$

$\therefore \boxed{f_T = 3}$

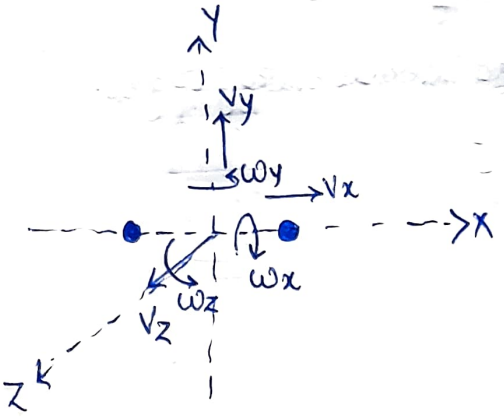
Rotational KE $= \frac{1}{2}I_x\omega_x^2 + \frac{1}{2}I_y\omega_y^2 + \frac{1}{2}I_z\omega_z^2$ (each $\omega^2 = 0$)

{As axis passes through molecule ⇒ I=0}

$\boxed{f_R = 0}$

$\therefore f_{monoatomic} = f_T + f_R = 3 + 0 = 3$

② Diatomic: Ex O_2, N_2, Cl_2



$TKE = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2$

$f_T = 3$

$RKE = \frac{1}{2}I_x\omega_x^2 + \frac{1}{2}I_y\omega_y^2 + \frac{1}{2}I_z\omega_z^2$ ($\omega_x^2 = 0$)

$f_R = 2$

$\therefore \boxed{f_{diatomic} = 5}$

Special case:

If T is very high ≈ 2000K

$f_{vibration} = 2$

$\therefore \boxed{f_{diatomic} = 7}$

③ Polyatomic

→ linear polyatomic: ex CO_2 (O=C=O) $f = f_T + f_R = 3 + 2 = 5$

→ Non-linear: ex H_2O $f = f_T + f_R = 3 + 3 = 6$

*If it is not stated, linear or non-linear, we take it as non-linear.

Law of Equipartition:

→ It states that a molecule on an average have equal energy corresponding to each degree of freedom and it will be $\frac{1}{2}kT$

k = Boltzmann const.
$k = \frac{R}{N_A} = 1.38 \times 10^{-23}$
T = temp in kelvin

1 molecule ⇒ 1 degree of freedom ⇒ $E = \frac{1}{2}kT$

1 molecule ⇒ f degree of freedom ⇒ $E = \frac{f}{2}kT$

1 mole molecule ⇒ f degree of freedom ⇒ $\left(\frac{f}{2}kT\right)N_A = \frac{f}{2}RT$

n mole molecule ⇒ f degree of freedom ⇒ $\boxed{E = \frac{f}{2}nRT = \frac{f}{2}PV}$

∴ Gas energy = KE + PE = $\frac{f}{2}nRT + 0$

$U = \frac{f}{2}nRT = \frac{f}{2}PV$ = Internal energy of gas

Monoatomic	$U = \frac{3}{2}nRT$	
Diatomic	$U = \frac{5}{2}nRT$	$= \underbrace{\frac{3}{2}nRT}_{\text{TKE}} + \underbrace{\frac{2}{2}nRT}_{\text{RKE}}$
Polyatomic (non-linear)	$U = 2nRT$	$= \frac{3}{2}nRT + \frac{3}{2}nRT$

Q A container have 64gm of O_2 gas at 27°C. Find internal energy of gas.

$$U = \frac{f}{2}nRT = \frac{5}{2} \times 2 \times \frac{25}{3} \times 300 = 12500\,J$$

$$n = \frac{64\,gm}{32\,gm/mol} = 2\,moles$$

Q A container (5lt) have He gas at 3atm pressure. Find its internal energy.

$$U = \frac{3}{2}PV = \frac{3}{2} \times 3 \times 1.01 \times 10^5 \times 5 \times 10^{-3} = 2272.5\,J$$